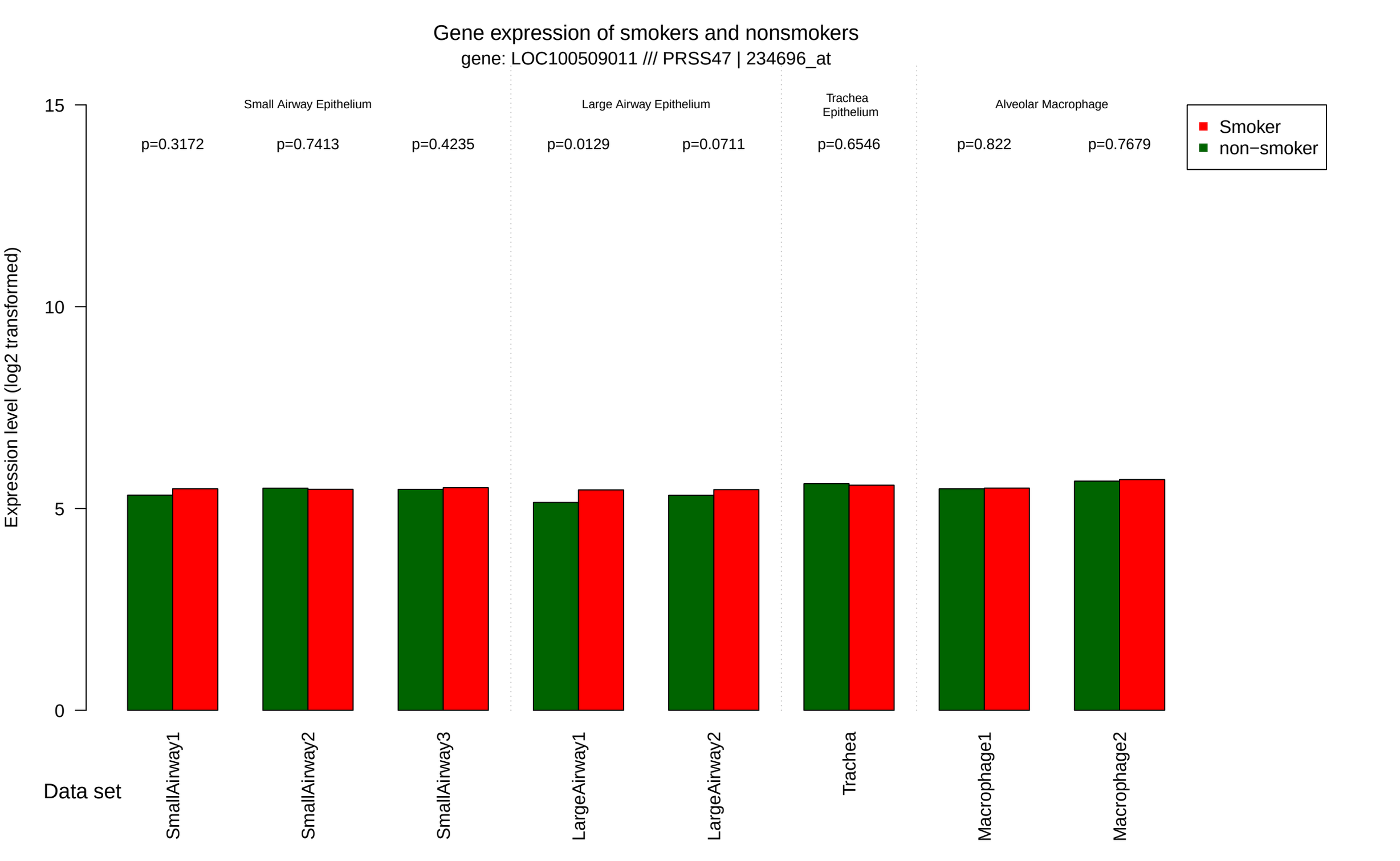
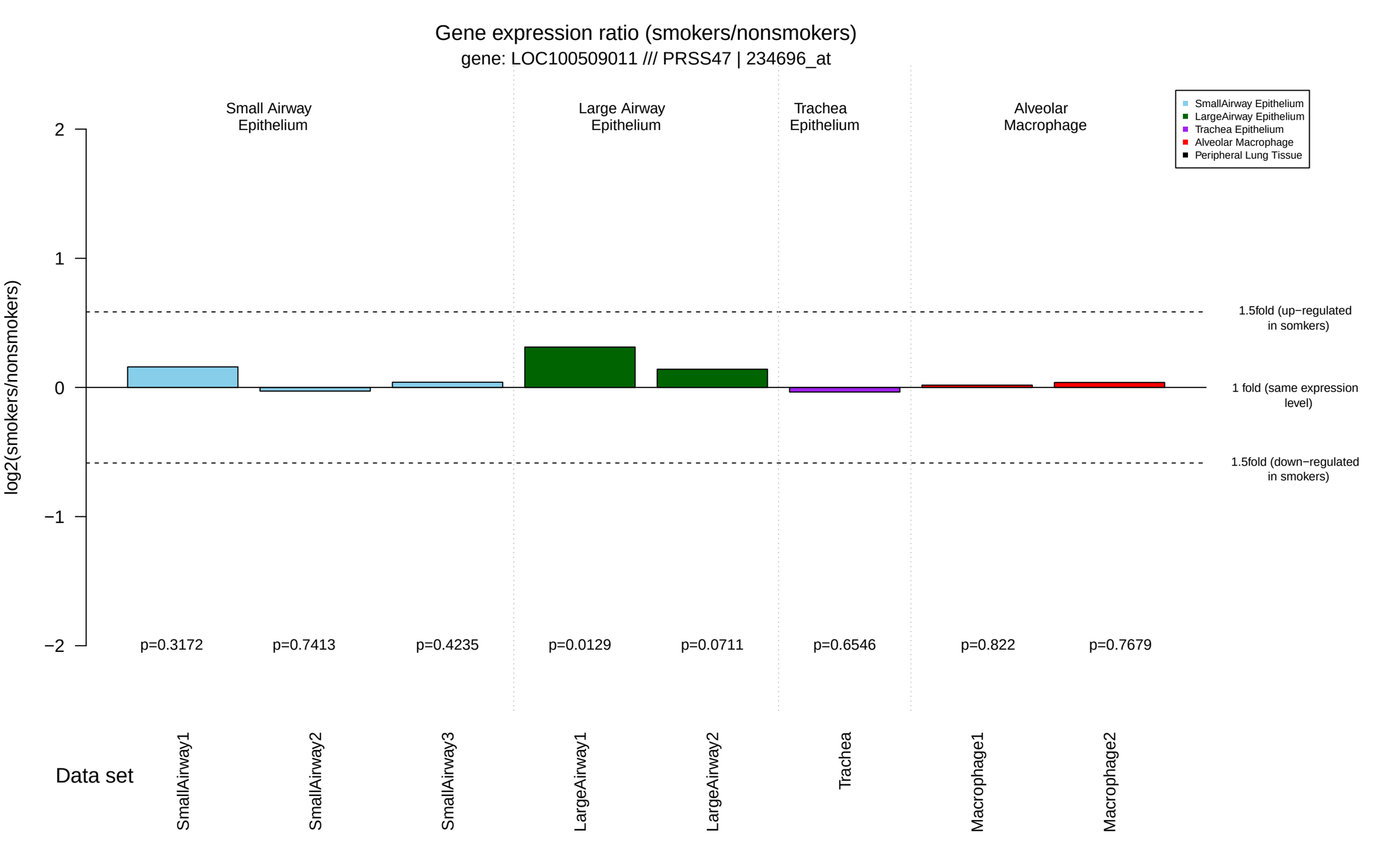
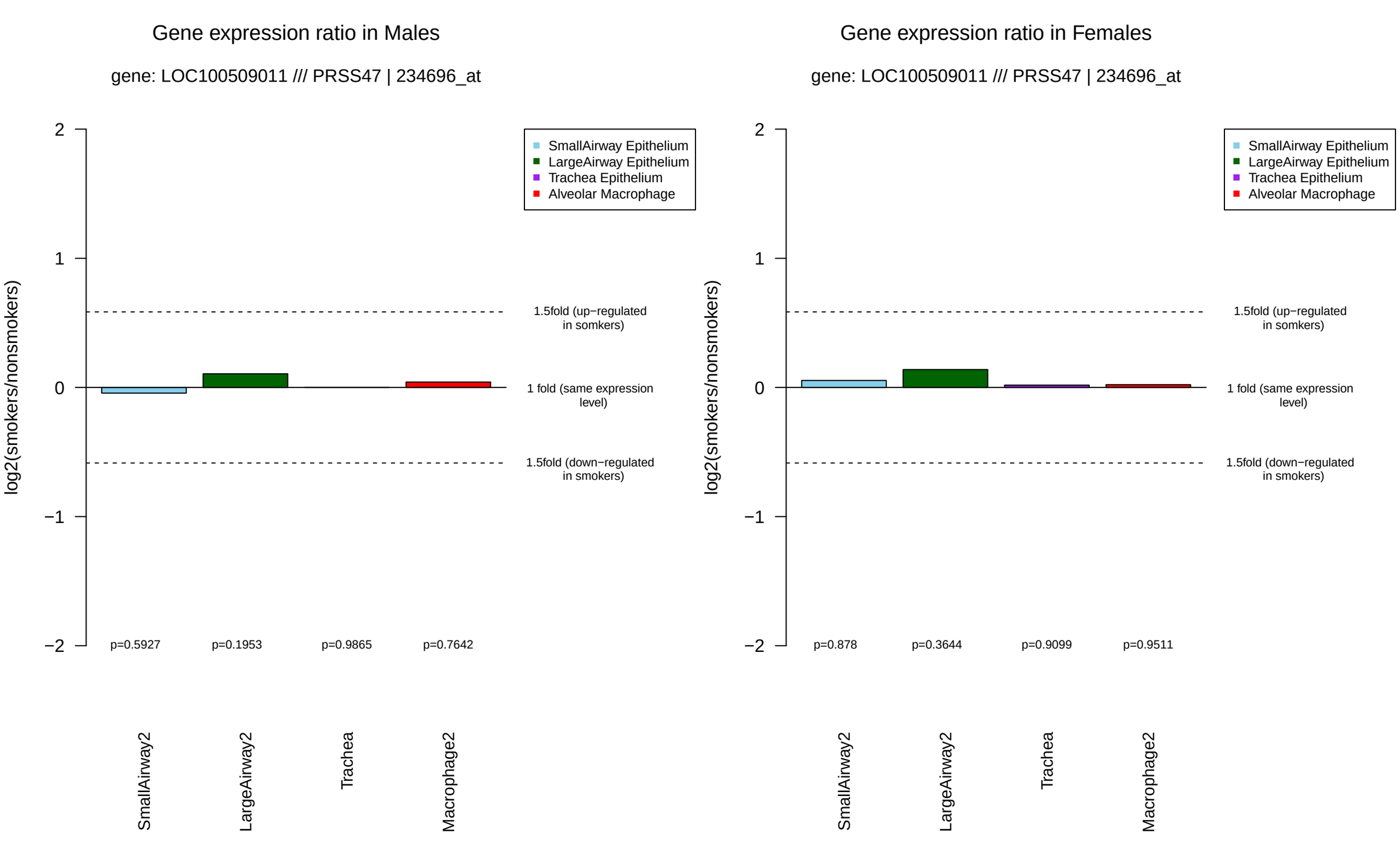
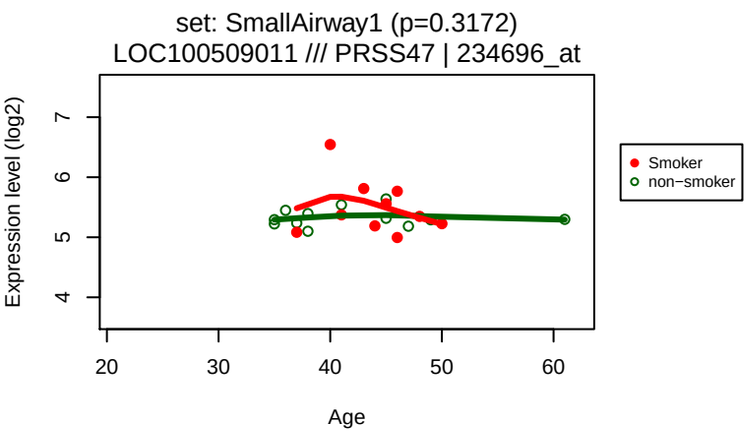
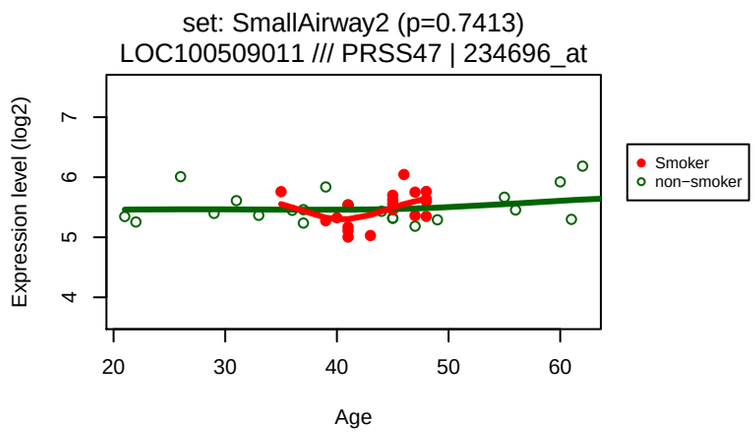
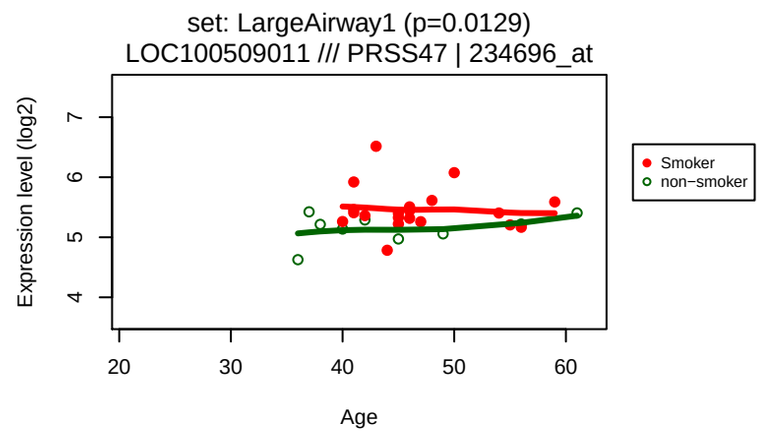

Gene expression of smokers and nonsmokers
gene: LOC100509011 /// PRSS47 | 234696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3172
p=0.7413
p=0.4235
p=0.0129
p=0.0711
p=0.6546
p=0.822
p=0.7679
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

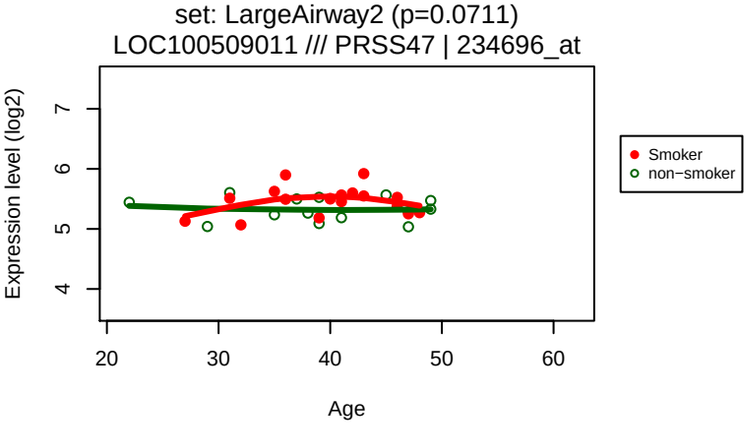

Gene expression ratio (smokers/nonsmokers)
gene: LOC100509011 /// PRSS47 | 234696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3172
p=0.7413
p=0.4235
p=0.0129
p=0.0711
p=0.6546
p=0.822
p=0.7679
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

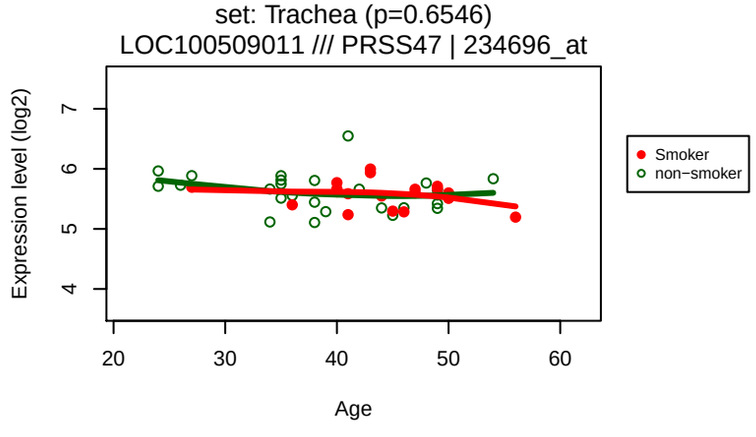

Gene expression ratio in Males
gene: LOC100509011 /// PRSS47 | 234696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5927
p=0.1953
p=0.9865
p=0.7642
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100509011 /// PRSS47 | 234696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.878
p=0.3644
p=0.9099
p=0.9511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

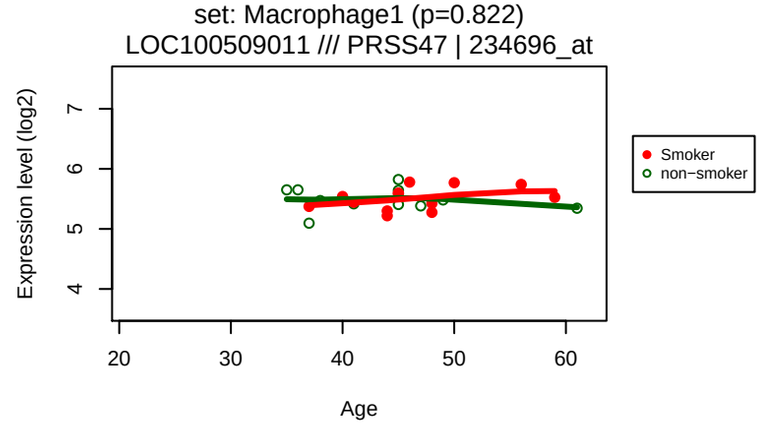

set: SmallAirway1 (p=0.3172)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker

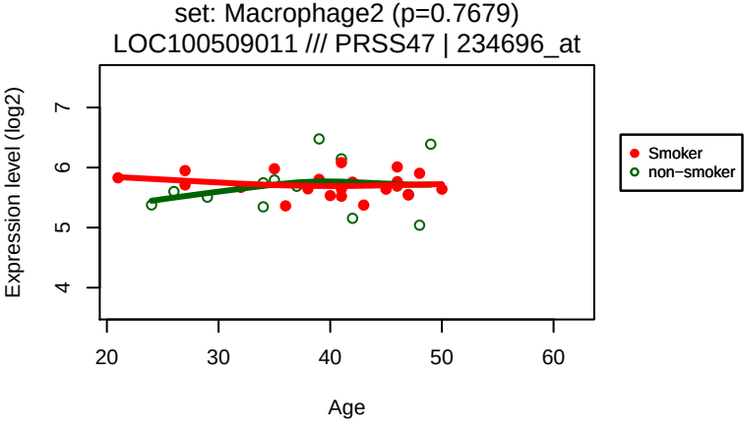

set: SmallAirway2 (p=0.7413)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker

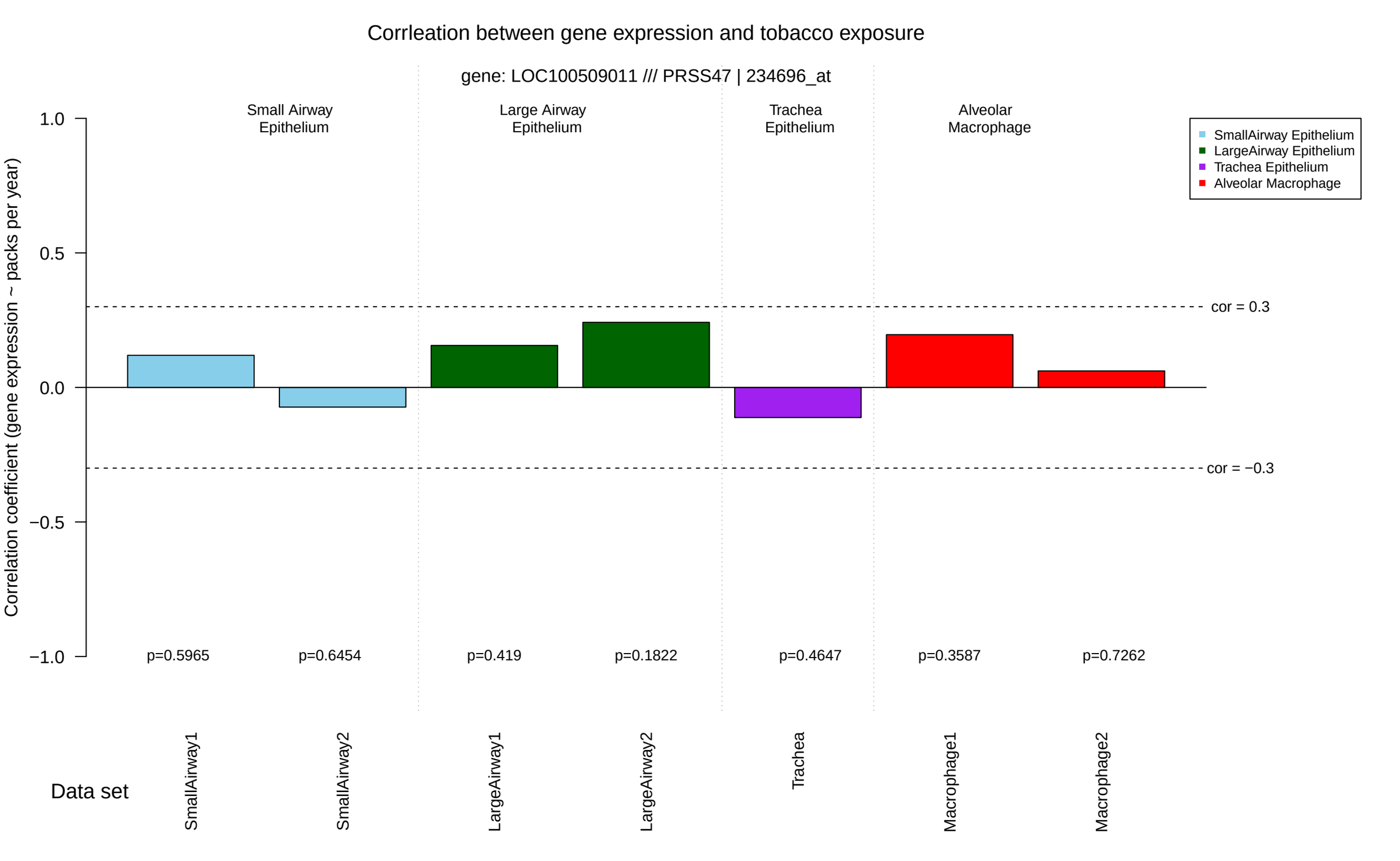

set: LargeAirway1 (p=0.0129)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0711)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6546)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.822)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7679)
LOC100509011 /// PRSS47 | 234696_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100509011 /// PRSS47 | 234696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5965
p=0.6454
p=0.419
p=0.1822
p=0.4647
p=0.3587
p=0.7262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2